

Chandar Rath

USA | +1 (562) 455 7838 | chandarrathi07@gmail.com | [LinkedIn](#) | [Github](#)

SUMMARY

Full-stack AI engineer who builds LLM products end-to-end and runs them in production — from RAG pipelines, agents, and APIs to the cloud infrastructure that ships and scales them. 4+ years across AI application development and DevOps, with hands-on work in LangChain, LangGraph, multi-provider LLM orchestration, vector retrieval, and evaluation, on AWS and GCP with Docker and Kubernetes. Comfortable owning an AI feature from first prompt to production SLA, under real latency, cost, and safety constraints.

TECHNICAL SKILLS

AI & LLM: LangChain, LangGraph, RAG (hybrid retrieval, reranking, embeddings), pgvector/FAISS, multi-provider LLMs (OpenAI, Anthropic, Bedrock), tool-calling, MCP, prompt engineering, guardrails, evaluation (Ragas, LangSmith)

Programming & Scripting: Python, Bash, YAML, JSON, SQL

Cloud & DevOps: AWS, GCP, Azure, Docker, Kubernetes, Terraform, CI/CD (GitHub Actions, Jenkins), Helm

Observability & Security: Prometheus, Grafana, Datadog, OpenTelemetry; IAM/VPC, OAuth2/JWT, PII redaction

Web & Application Development: React, Next.js, Node.js, FastAPI, Flask, REST, GraphQL

Databases: MongoDB, MySQL, PostgreSQL, Redis, DynamoDB

Architecture: Microservices, Containerization, Cloud-Native Applications, Serverless Architecture, Event-Driven Systems, Load Balancing, High Availability

Security & Compliance: IAM Policies, Secrets Manager, VPC Security Groups, SSL/TLS, API Authentication (OAuth2, JWT), Vulnerability Scanning

PROFESSIONAL EXPERIENCE

Citi, USA

Mar 2025 – Present

Full Stack AI Engineer

- Built a production RAG assistant that lets relationship managers pull answers from dense policy and product documents instead of searching manually, improving answer relevance by 27% using semantic chunking, embeddings, and hybrid retrieval (BM25 + vector) with cross-encoder reranking over a pgvector store.
- Owned the retrieval and serving layers on a small team, developing agentic workflows in LangGraph with tool-calling and guardrails so the assistant could complete multi-step servicing tasks safely and auditably.
- Cut inference cost by 40% by routing requests across multiple LLM providers (OpenAI, Anthropic, Bedrock) by query complexity, with fallback and prompt caching, while serving on Kubernetes with autoscaling to hold latency steady under load.
- Built a Ragas/LangSmith evaluation harness wired into CI that scored faithfulness, context relevance, and recall on every deploy, catching most quality regressions before release, with tracing through OpenTelemetry.
- Hardened the system for a regulated environment by layering prompt-injection defense, PII redaction, structured-output schemas, and IAM/VPC/secrets controls into the request path.

HCL, India

Aug 2022 – Dec 2023

Software Engineer

- Rebuilt the front end of enterprise web applications in React.js and Node.js/Express.js around a reusable component architecture, cutting page-load times by 25% and improving user engagement.
- Designed REST and GraphQL APIs over MongoDB and PostgreSQL, adding caching and query tuning to support data-heavy, real-time features.
- Containerized services with Docker and Kubernetes and automated CI/CD with Jenkins, GitHub Actions, and Terraform, reducing failed deployments by 35% and shortening release cycles.
- Managed AWS deployments (EC2, S3, RDS) and set up monitoring with Prometheus, Grafana, and CloudWatch to keep production stable and surface issues early.

PROJECTS

RAG Ops — Production RAG Assistant · LangChain · pgvector · FastAPI · React

- Built and open-sourced a from-scratch RAG assistant to benchmark retrieval strategies, publishing a reproducible naive-vs-hybrid-plus-reranking comparison that moved NDCG@5 from 0.64 to 0.89 on a 500-question eval set.
- Served quantized open models (Llama 3 / Mistral) on a single GPU with vLLM, keeping per-query cost near zero and P95 latency under 800ms.
- Shipped a public demo with a FastAPI backend, a streaming React/Next.js UI, and a Ragas dashboard exposing faithfulness and recall per query.

AskOps — Agentic AI Assistant · LangGraph · MCP · Multi-provider · K8s

- Built a tool-calling agent that resolves multi-step tasks from natural language by pulling from internal APIs, docs, and tickets, taking a workflow that ran 15 minutes per request down to under a minute.
- Designed the agent loop in LangGraph with MCP tool integration and guardrails — scoped tools and human-in-the-loop on irreversible actions — so the agent could act without acting unsafely.
- Ran it reliably with provider fallback, prompt caching, and autoscaled serving on Kubernetes, with full request tracing through LangSmith.

SLM Fine-Tuning Pipeline — Domain-Specialized Small Model · QLoRA · Llama 3 8B · vLLM · FastAPI

- Fine-tuned Llama 3 8B with QLoRA to match a GPT-4-class model on a narrow domain task at a fraction of the cost, lifting task accuracy from ~71% base to ~89% on a held-out set.
- Kept it reproducible on a single 24 GB consumer GPU by 4-bit quantizing and training only LoRA adapters, then merging and serving with vLLM.
- Documented the full dataset-prep → train → eval → deploy pipeline behind a FastAPI endpoint and a Gradio demo so anyone can re-run it.

LLM Observability & Cost Platform — Self-Hosted Tracing for AI Apps · OpenTelemetry · ClickHouse · Grafana · FastAPI

- Built a self-hostable observability layer that traces every LLM request end to end — prompt, retrieved context, tool calls, tokens, latency, and cost — so teams can debug and budget AI systems instead of guessing.
- Cut blind spots in production AI by storing traces in ClickHouse and surfacing per-route cost and latency in Grafana, with automated alerts when spend or quality drifts.
- Shipped it as drop-in middleware that instruments any LangChain or LangGraph app in a few lines, with a public demo and docs.

EDUCATION

Master of Science in Computer Science (GPA: 3.7/4)

California State University, USA

Bachelor of Engineering in Computer Science (GPA: 8.9/ 10)

Gujarat Technological University, India